# Laboratório 7 - SQLite Parte II (UW-Madison)

Pedro Monteiro Tilhe - RA 254672

2026-09-22

```
library(DBI)
```

```
Warning: pacote 'DBI' foi compilado no R versão 4.5.3
```

```
library(RSQLite)
```

```
Warning: pacote 'RSQLite' foi compilado no R versão 4.5.3
```

```
conn <- dbConnect(SQLite(), "datasets/uwmadison.sqlite3")
knitr::opts_chunk$set(connection = "conn")
```

### 1. Conexão com o banco de dados

```
dbListTables(conn)
```

```
 [1] "course_offerings"    "courses"             "grade_distributions"
 [4] "instructors"         "rooms"               "schedules"
 [7] "sections"            "subject_memberships" "subjects"
[10] "teachings"
```

### 2. Professores que lecionaram disciplinas de Estatística

```sql
SELECT COUNT(DISTINCT i.id) AS N_PROFESSORES
FROM subjects s
INNER JOIN subject_memberships sm ON sm.subject_code = s.code
INNER JOIN sections sec ON sec.course_offering_uuid = sm.course_offering_uuid
INNER JOIN teachings t ON t.section_uuid = sec.uuid
INNER JOIN instructors i ON i.id = t.instructor_id
WHERE s.abbreviation = 'STAT';
```

Table 1: 1 records

| N_PROFESSORES |
|---|
| 379 |

```sql
SELECT DISTINCT i.id AS ID, i.name AS PROFESSOR
FROM subjects s
INNER JOIN subject_memberships sm ON sm.subject_code = s.code
INNER JOIN sections sec ON sec.course_offering_uuid = sm.course_offering_uuid
INNER JOIN teachings t ON t.section_uuid = sec.uuid
INNER JOIN instructors i ON i.id = t.instructor_id
WHERE s.abbreviation = 'STAT'
ORDER BY i.name;
```

Table 2: Displaying records 1 - 10

| ID | PROFESSOR |
|---|---|
| 4787911 | ABIGAIL BENZINE |
| 3111842 | ADIN-CRISTIAN ANDREI |
| 3775809 | AKICHIKA OZEKI |
| 4958796 | ALAN HUANG |
| 5451706 | ALBERTO DEL PIA |
| 3661782 | ALBRECHT KLEMM |
| 5656034 | ALEXANDER COVINGTON |
| 4680082 | ALEXANDER FISH |
| 3573775 | ALEXANDER KISELEV |
| 5162339 | ALYSSA DIGILIO |

### 3. Nota média (GPA) de cada oferecimento

```sql
SELECT gd.course_offering_uuid AS OFERECIMENTO,
       SUM(gd.a_count*4.0 + gd.ab_count*3.5 + gd.b_count*3.0 + gd.bc_count*2.5
           + gd.c_count*2.0 + gd.d_count*1.0 + gd.f_count*0.0)
       / SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count
             + gd.c_count + gd.d_count + gd.f_count) AS GPA
FROM grade_distributions gd
INNER JOIN subject_memberships sm
        ON sm.course_offering_uuid = gd.course_offering_uuid
INNER JOIN subjects s ON s.code = sm.subject_code
WHERE s.abbreviation = 'STAT'
GROUP BY gd.course_offering_uuid
HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count
           + gd.c_count + gd.d_count + gd.f_count) > 0
ORDER BY GPA;
```

Table 3: Displaying records 1 - 10

| OFERECIMENTO | GPA |
|---|---|
| f5ed01ad-4025-3818-9391-ad4bb97f6607 | 2.441177 |
| 6c66b19a-c860-3d01-8114-bec787ff2409 | 2.607692 |
| f5528bfe-6249-39e7-912a-761f3c1e483e | 2.618182 |
| 413d320f-9979-3458-826b-3ad2f317313c | 2.633803 |
| f6808fed-2ffc-3d10-bf8e-4f902f5927d2 | 2.642857 |
| c49ccf8c-4a95-3b24-8560-b581251c138d | 2.659420 |
| 05712bf4-5571-3967-894d-3af34280012e | 2.680328 |
| d2fae0ac-24bf-33f4-a9aa-e12ea0df30f4 | 2.683823 |
| 20300181-aa5e-34f5-becf-7631d8c6d45c | 2.690790 |
| 045adab4-b4ea-3831-815e-20486d2e7d33 | 2.701754 |

### Professor mais difícil

```sql
WITH stat_off AS (
  SELECT DISTINCT sm.course_offering_uuid AS off_uuid
  FROM subject_memberships sm
  INNER JOIN subjects s ON s.code = sm.subject_code
  WHERE s.abbreviation = 'STAT'
),
gpa_off AS (
  SELECT gd.course_offering_uuid AS off_uuid,
         SUM(gd.a_count*4.0 + gd.ab_count*3.5 + gd.b_count*3.0 + gd.bc_count*2.5
```

```
            + gd.c_count*2.0 + gd.d_count*1.0 + gd.f_count*0.0)
        / SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count
              + gd.c_count + gd.d_count + gd.f_count) AS gpa
  FROM grade_distributions gd
  INNER JOIN stat_off so ON so.off_uuid = gd.course_offering_uuid
  GROUP BY gd.course_offering_uuid
  HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count
            + gd.c_count + gd.d_count + gd.f_count) > 0
),
prof_off AS (
  SELECT DISTINCT t.instructor_id AS iid, sec.course_offering_uuid AS off_uuid
  FROM teachings t
  INNER JOIN sections sec ON sec.uuid = t.section_uuid
  INNER JOIN stat_off so ON so.off_uuid = sec.course_offering_uuid
)
SELECT i.name AS PROFESSOR, AVG(g.gpa) AS GPA_MEDIO
FROM prof_off p
INNER JOIN gpa_off g ON g.off_uuid = p.off_uuid
INNER JOIN instructors i ON i.id = p.iid
GROUP BY p.iid, i.name
ORDER BY GPA_MEDIO ASC
LIMIT 1;
```

Table 4: 1 records

| PROFESSOR | GPA_MEDIO |
|---|---|
| JAMES D KUELBS | 2.607692 |

**Professor mais fácil**

```
WITH stat_off AS (
  SELECT DISTINCT sm.course_offering_uuid AS off_uuid
  FROM subject_memberships sm
  INNER JOIN subjects s ON s.code = sm.subject_code
  WHERE s.abbreviation = 'STAT'
),
gpa_off AS (
  SELECT gd.course_offering_uuid AS off_uuid,
         SUM(gd.a_count*4.0 + gd.ab_count*3.5 + gd.b_count*3.0 + gd.bc_count*2.5
             + gd.c_count*2.0 + gd.d_count*1.0 + gd.f_count*0.0)
```

```
        / SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count
             + gd.c_count + gd.d_count + gd.f_count) AS gpa
  FROM grade_distributions gd
  INNER JOIN stat_off so ON so.off_uuid = gd.course_offering_uuid
  GROUP BY gd.course_offering_uuid
  HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count
           + gd.c_count + gd.d_count + gd.f_count) > 0
),
prof_off AS (
  SELECT DISTINCT t.instructor_id AS iid, sec.course_offering_uuid AS off_uuid
  FROM teachings t
  INNER JOIN sections sec ON sec.uuid = t.section_uuid
  INNER JOIN stat_off so ON so.off_uuid = sec.course_offering_uuid
)
SELECT i.name AS PROFESSOR, AVG(g.gpa) AS GPA_MEDIO
FROM prof_off p
INNER JOIN gpa_off g ON g.off_uuid = p.off_uuid
INNER JOIN instructors i ON i.id = p.iid
GROUP BY p.iid, i.name
HAVING COUNT(*) >= 5
ORDER BY GPA_MEDIO DESC
LIMIT 1;
```

Table 5: 1 records

| PROFESSOR | GPA_MEDIO |
|---|---|
| GRACE WAHBA | 3.972642 |

### Disciplina mais difícil

```
WITH stat_off AS (
  SELECT DISTINCT sm.course_offering_uuid AS off_uuid
  FROM subject_memberships sm
  INNER JOIN subjects s ON s.code = sm.subject_code
  WHERE s.abbreviation = 'STAT'
),
gpa_off AS (
  SELECT gd.course_offering_uuid AS off_uuid,
         SUM(gd.a_count*4.0 + gd.ab_count*3.5 + gd.b_count*3.0 + gd.bc_count*2.5
             + gd.c_count*2.0 + gd.d_count*1.0 + gd.f_count*0.0)
```

```
        / SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count
              + gd.c_count + gd.d_count + gd.f_count) AS gpa
  FROM grade_distributions gd
  INNER JOIN stat_off so ON so.off_uuid = gd.course_offering_uuid
  GROUP BY gd.course_offering_uuid
  HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count
             + gd.c_count + gd.d_count + gd.f_count) > 0
)
SELECT c.number AS NUMERO, c.name AS DISCIPLINA, AVG(g.gpa) AS GPA_MEDIO
FROM gpa_off g
INNER JOIN course_offerings co ON co.uuid = g.off_uuid
INNER JOIN courses c ON c.uuid = co.course_uuid
GROUP BY c.uuid, c.number, c.name
ORDER BY GPA_MEDIO ASC
LIMIT 1;
```

Table 6: 1 records

| NUMERO | DISCIPLINA | GPA_MEDIO |
|---|---|---|
| 431 | Introduction to the Theory of Probability | 2.87807 |

### Disciplina mais fácil

```
WITH stat_off AS (
  SELECT DISTINCT sm.course_offering_uuid AS off_uuid
  FROM subject_memberships sm
  INNER JOIN subjects s ON s.code = sm.subject_code
  WHERE s.abbreviation = 'STAT'
),
gpa_off AS (
  SELECT gd.course_offering_uuid AS off_uuid,
         SUM(gd.a_count*4.0 + gd.ab_count*3.5 + gd.b_count*3.0 + gd.bc_count*2.5
             + gd.c_count*2.0 + gd.d_count*1.0 + gd.f_count*0.0)
         / SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count
               + gd.c_count + gd.d_count + gd.f_count) AS gpa
  FROM grade_distributions gd
  INNER JOIN stat_off so ON so.off_uuid = gd.course_offering_uuid
  GROUP BY gd.course_offering_uuid
  HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count
             + gd.c_count + gd.d_count + gd.f_count) > 0
```

```
)
SELECT c.number AS NUMERO, c.name AS DISCIPLINA, AVG(g.gpa) AS GPA_MEDIO
FROM gpa_off g
INNER JOIN course_offerings co ON co.uuid = g.off_uuid
INNER JOIN courses c ON c.uuid = co.course_uuid
GROUP BY c.uuid, c.number, c.name
HAVING COUNT(*) >= 5
ORDER BY GPA_MEDIO DESC
LIMIT 1;
```

Table 7: 1 records

| NUMERO | DISCIPLINA | GPA_MEDIO |
|---|---|---|
| 860 | Estimation of Functions from Data | 3.977381 |

### 4. Desconectar do banco de dados

```
dbDisconnect(conn)
```

## Data e hora de compilação

```
[1] "22/09/2026 17:08:26"
```